Health-promoting behaviours and concussion history are associated with cognitive function, mood-related symptoms and emotional–behavioural dyscontrol in former NFL players: an NFL-LONG Study

Samuel R Walton ,[1] Zachary Y Kerr ,[1] Benjamin L Brett ,[2] Avinash Chandran,[1,3] J D DeFreese,[1] Abbie E Smith-Ryan,[1] Lee Stoner,[1] Ruben J Echemendia ,[4,5] Michael McCrea ,[2] William P Meehan, III,[6,7] Kevin M Guskiewicz[1]

► Additional material is published online only. To view, please visit the journal online (http://dx.doi.org/10.1136/bjsports-2020-103400).

[1]Exercise and Sport Science, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina, USA
[2]Neurosurgery/Neurology, Medical College of Wisconsin, Milwaukee, Wisconsin, USA
[3]NCAA Injury Surveillance Program, Datalys Center for Sports Injury Research and Prevention, Indianapolis, Indiana, USA
[4]Psychology, University of Missouri Kansas City, Kansas City, Missouri, USA
[5]Neuropsychology, University Orthopedics Center Concussion Clinic, State College, PA, USA
[6]Sports Medicine, Boston Children's Hospital, Boston, Massachusetts, USA
[7]Division of Emergency Medicine, Boston Children's Hospital, Boston, Massachusetts, USA

Correspondence to
Dr Zachary Y Kerr, Exercise and Sport Science, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina, USA; zkerr@email.unc.edu

Accepted 16 December 2020

Check for updates

© Author(s) (or their employer(s)) 2021. No commercial re-use. See rights and permissions. Published by BMJ.

To cite: Walton SR, Kerr ZY, Brett BL, *et al*. *Br J Sports Med* Epub ahead of print: [*please include* Day Month Year]. doi:10.1136/bjsports-2020-103400

ABSTRACT

Objectives To examine the relationships among self-reported sport-related concussion (SRC) history and current health-promoting behaviours (exercise frequency, diet quality and sleep duration) with self-reported measures of brain health (cognitive function, symptoms of depression and anxiety and emotional–behavioural dyscontrol) in former NFL players.

Methods In this cross-sectional study, a questionnaire was sent to former NFL players. Respondents reported SRC history (categorical: 0; 1–2; 3–5; 6–9; 10+ concussions), number of moderate-to-vigorous aerobic and resistance exercise sessions per week, diet quality (Rapid Eating Assessment for Participants—Shortened) and average nightly sleep duration. Outcomes were Patient-Reported Outcomes Measurement Information System Cognitive Function, Depression, and Anxiety, and Neuro-QoL Emotional-Behavioral Dyscontrol domain T-scores. Multivariable linear regression models were fit for each outcome with SRC history, exercise frequency, diet quality and sleep duration as explanatory variables alongside select covariates.

Results Multivariable regression models (n=1784) explained approximately 33%–38% of the variance in each outcome. For all outcomes, SRC history ($0.144 \leq |\beta| \leq 0.217$) was associated with poorer functioning, while exercise frequency ($0.064 \leq |\beta| \leq 0.088$) and diet quality ($0.057 \leq |\beta| \leq 0.086$) were associated with better functioning. Sleeping under 6 hours per night ($0.061 \leq |\beta| \leq 0.093$) was associated with worse depressive symptoms, anxiety and emotional–behavioural dyscontrol.

Conclusion Several variables appear to be associated with mood and perceived cognitive function in former NFL players. SRC history is non-modifiable in former athletes; however, the effects of increasing postplaying career exercise frequency, making dietary improvements, and obtaining adequate sleep represent important potential opportunities for preventative and therapeutic interventions.

Sport-related concussion (SRC) is a common injury in collision sports, including American football,[1 2] yet the neurobiological understanding of the long-term effects of SRC on brain health is unknown.[3] Studies of former professional and collegiate American football players have suggested that repetitive SRCs might negatively affect long-term cognitive function,[4–13] mood-related symptoms and disorders (eg, depression)[4 10 14–19] and executive functioning (eg, disinhibition).[4 14 20–25] A limitation of previous research is the lack of consideration of factors that influence the relationship between SRC history and brain health. Specifically, studies of health-promoting factors related to neurobehavioural outcomes are sparse in this population.[26 27] Although personal history of SRCs is not modifiable, there may be modifiable factors associated with long-term health that could be leveraged to improve function and quality of life.

Much of the growing body of research in former collision-sport athletes examines factors relating to poor functioning (eg, mood-related symptoms) and risk of neurodegenerative disease (eg, mild cognitive impairment). Interrogation of protective factors that positively affect long-term health and quality of life (ie, exercise, diet and sleep behaviours) may address key knowledge gaps and open avenues for promoting brain health in these individuals. Specifically in former American football players, self-reported physical inactivity, changes in appetite, lower diet quality, sleep disturbance and fatigue have all been associated with self-reported cognitive difficulties and emotional distress.[10 28–35] Beyond former athletes, exercise, diet and sleep are potentially modifiable health-promoting behaviours that have previously been shown to improve cognitive function,[26 36–41] mood-related symptoms[26 40 42] and general health-related quality of life in older individuals.[26 43 44] Moderate-to-vigorous exercise performed two or more times per week over multiple years has been linked to the attenuation of cognitive decline in typical ageing, preservation of brain structure and slowing of neurodegenerative disease progression in older adults.[36 45] Further, atypical nighttime sleep (ie, less than 6 or greater than 9 hours per night) has been associated with increased brain atrophy, worse cognitive function and a faster cognitive decline with ageing.[37 38 46 47] Thus, efforts to modify exercise, diet and sleep may be preventative and therapeutic targets for clinical intervention in former athletes.

Br J Sports Med: first published as 10.1136/bjsports-2020-103400 on 4 January 2021. Downloaded from http://bjsm.bmj.com/ on January 5, 2021 at Nationwide Children's Hospital. Protected by copyright.

The present study examined the associations of self-reported SRC history and health-promoting behaviours (exercise frequency, diet quality and sleep duration) with self-reported measures of cognition and mood (cognitive function, symptoms of depression and anxiety, and emotional–behavioural dyscontrol) in former NFL players. We hypothesised that: (1) more self-reported lifetime SRCs would be associated with worse cognitive function, greater mood-related symptom burden and more emotional–behavioural dyscontrol; and (2) more frequent exercise, better diet quality and adequate sleep (eg, between 6 and 9 hours per night)[46 47] would be associated with better cognitive function, lower mood-related symptom burden and less emotional–behavioural dyscontrol.

METHODS

This cross-sectional study surveyed former NFL players using a self-administered questionnaire that was provided either online (Qualtrics, SAP America, Newton Square, Pennsylvania, USA) or via paper hardcopy. The study was approved by the Institutional Review Board at the University of North Carolina at Chapel Hill. All participants provided written informed consent prior to participation. This was required in order to gain access to the electronic survey; or it was completed on the first page of the hardcopy survey, prior to completing any demographic or study-related items.

Participants

Former NFL players were contacted between 1 January 2019 and 15 February 2020 through email and hard-copy letters mailed to home addresses through existing relationships with the NFL Alumni Association, as well as individual teams and players (figure 1). Participants were required to have played at least one season in the NFL to be eligible for the study.

Data collection

The current study was part of an ongoing project entitled, *Neurologic Function across the Lifespan: A Prospective, LONGitudinal, and Translational Study for Former National Football League Players (NFL-LONG)*. The questionnaire captured demographic information, football playing history, medical history, concussion history, musculoskeletal injury history, health status, nutrition, substance use and current health-related habits (table 1).

Patient involvement

The comprehensive questionnaire was a modified version of a previously administered questionnaire—enhanced with input from epidemiologists, athletic trainers, neuropsychologists, physicians and other sports medicine professionals.[8 15] Additionally, five former NFL players provided critical feedback to improve the questionnaire's clarity and flow.

Outcome measures of self-reported functioning

Main outcome measures were the PROMIS Adult Cognitive Function Short Form-4a (COG),[48] the four-item Depressive Symptoms (DEP) and Anxiety (ANX) scales from the PROMIS-29 Profile V.2.0[49] and the Emotional and Behavioral Dyscontrol section of the Neuro-QoL battery (EBD).[50 51] These measures were developed with support from the National Institutes of Health in the USA using classical test and item-response theories to concisely and precisely measure their intended constructs.[48–51] All measures consisted of items that were rated

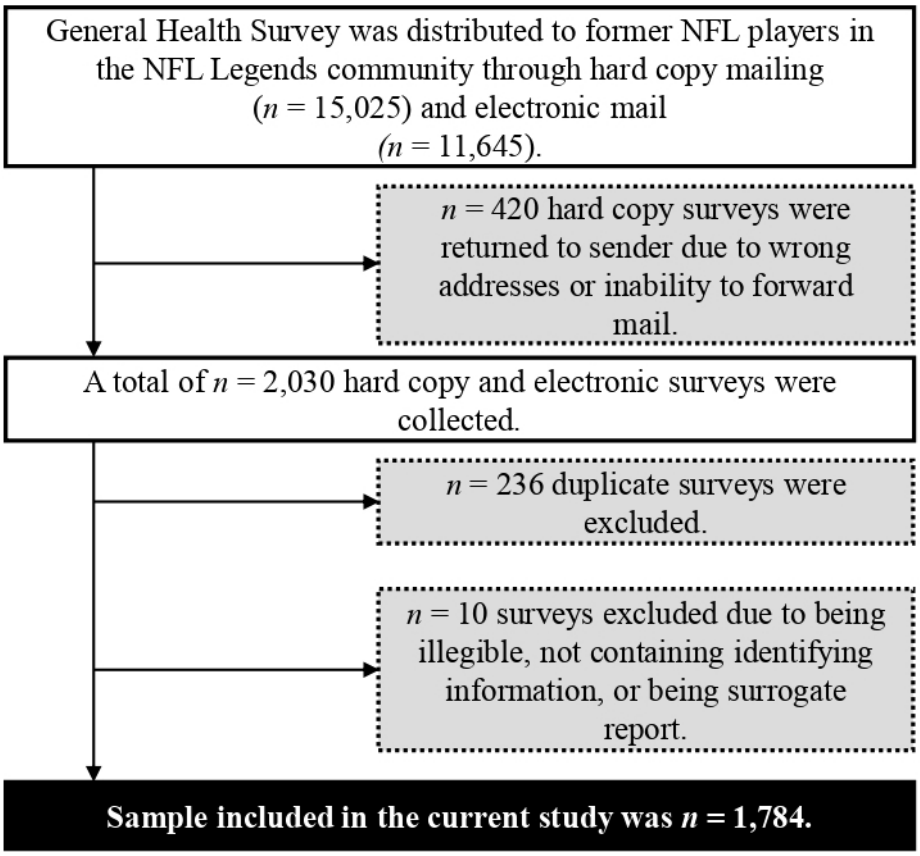

Figure 1 Participant recruitment and general health survey collection.

Br J Sports Med: first published as 10.1136/bjsports-2020-103400 on 4 January 2021. Downloaded from http://bjsm.bmj.com/ on January 5, 2021 at Nationwide Children's Hospital. Protected by copyright.

Table 1 Sample characteristics of former NFL players (n=1784)

	Full sample with no imputed data n=1784	Full sample after multiple imputation (pooled) n=1784
Age*, mean (SD)	52.3 (16.3)	52.3 (16.3)
Missing, n (%)	0 (0.0)	–
Body mass index (BMI)†, mean (SD)	30.8 (4.6)	30.8 (4.6)
Missing, n (%)	17 (0.9)	–
Race/ethnicity, n (%)		
White/non-Hispanic	1041 (58.4)	1047 (58.7)
Person of colour	732 (41.0)	737 (41.3)
Missing, n (%)	11 (0.6)	–
Relationship status, n (%)		
In a relationship with a significant other	1430 (80.2)	1433 (80.3)
No significant other	350 (19.6)	351 (19.7)
Missing, n (%)	4 (0.2)	–
Highest education attained, n (%)		
Less than a Bachelors' degree (eg, associate, diploma)	268 (15.0)	268 (15.0)
Bachelor's degree (eg, BA, BS)	1089 (61.0)	1090 (61.1)
Postgraduate university degree (eg, MA, PhD, JD, MD)	425 (23.8)	426 (23.9)
Missing, n (%)	2 (0.1)	–
Total years of football played, mean (SD)	17.5 (4.5)	17.5 (4.5)
Missing, n (%)	59 (3.3)	–
Lifetime sport-related concussion history, n (%)		
0	234 (13.1)	234 (13.1)
1–2	313 (17.5)	313 (17.5)
3–5	425 (23.8)	425 (23.8)
6–9	366 (20.5)	366 (20.5)
10 or more	446 (25.0)	446 (25.0)
Missing, n (%)	0 (0.0)	–
Osteoarthritis/degenerative arthritis diagnosis, n (%)	330 (18.5)	330 (18.5)
Missing, n (%)	0 (0.0)	–
Illicit drug use in past 3 months, n (%)	354 (19.8)	354 (19.8)
Missing, n (%)	0 (0.0)	–
Frequency of alcohol use, n (%)		
Never	320 (17.9)	325 (18.2)
Monthly or less	351 (19.7)	358 (20.1)
2–4 times/month	464 (26.0)	474 (26.6)
2–3 times/week	350 (19.6)	354 (19.8)
4 or more times/week	270 (15.1)	273 (15.3)
Missing, n (%)	29 (1.6)	–
Current pain (PROMIS-29)‡, mean (SD)	4.1 (2.5)	4.1 (2.5)
Missing, n (%)	19 (1.1)	–
Exercise frequency§, mean (SD)	5.3 (3.3)	5.2 (3.3)
Missing, n (%)	307 (17.2)	–
REAPS sum score, mean (SD)	28.0 (4.3)	27.9 (4.3)
Missing, n (%)	205 (11.5)	–
Sleep duration		
Low sleep group (<6 hours/night), n (%)	155 (8.7)	169 (9.5)
Intermediate sleep group (6–9 hours/night), n (%)	1361 (76.3)	1426 (79.9)
High sleep group (>9 hours/night), n (%)	174 (9.8)	189 (10.6)
Missing, n (%)	94 (5.3)	–

Descriptive data are provided for participants in the current study both prior to imputation of missing data and afterward. The multiple imputation procedure is described in detail the supplement to this paper.
*Age was measured in years.
†BMI was measured in kg/m^2.
‡Pain was self-reported on a Likert Scale from 0 to 10 where 0 was 'no pain' and 10 was 'worst pain imaginable'.
§Exercise frequency is reported as the sum of the typical number of summed moderate-to-vigorous aerobic exercise and resistance training exercise sessions per week.
PROMIS, Patient-Reported Outcomes Measurement Information System; REAPS, Rapid Eating Assessment for Participants—Shortened Version.

on a 5-point Likert Scale (table 2). Scores for individual items were summed and transformed to T-scores for each domain such that a score of 50 represented the US population normative mean, with a SD of 10. Higher scores indicated a more adaptive outcome in the COG domain and less adaptive outcomes for DEP, ANX and EBD.

SRC history

Participants were prompted to report all concussions that they believed they had sustained while playing football, including instances that were not medically diagnosed. A definition of concussion, used in prior research, was provided to participants[52]: 'A concussion typically occurs from a blow to the head and is followed by a variety of symptoms that may include any of the following: headache, dizziness, loss of balance, blurred vision, 'seeing stars', feeling in a fog, or slowed down, memory problems, poor concentration, nausea, or throwing-up. Getting 'knocked out' or being unconscious does NOT always occur with a concussion'. This definition was selected to be consistent with prior studies that included many of the same participants and, while not exactly the same, the authors felt it was not disparate from the most recent international consensus definition from the concussion in sport group.[53] Respondents separately reported the number of SRCs they suspected they sustained on an ordinal scale (ie, 0, 1, 2, …, '10 or more') for each of the following categories: prior to high school, during high school, during college, as a professional football player and during 'other' (eg, military-related) football. Self-reported SRC was collapsed into five categories based on the sums reported across all levels: 0; 1–2; 3–5; 6–9; and 10+ lifetime SRCs.

Current health-promoting behaviours (exercise frequency, diet quality, sleep duration)

Current exercise participation was simply reported as the typical number of weekly sessions of moderate-to-vigorous aerobic exercise (defined as exercise that increases breathing rate and heart rate (eg, jogging, cycling)) or resistance exercise (defined as resistance training or weight lifting). No minimum session duration was indicated. Exercise frequency was the sum of moderate-to-vigorous exercise sessions per week, plus the number of resistance exercise sessions per week.

Current diet quality was assessed with the Rapid Eating Assessment for Participants—Shortened Version (REAPS).[54] The REAPS captures the types and frequency of food consumption (eg, whole grains, sweets), as well as the preparation of meals (eg, restaurant vs home). It is a brief, 16-item instrument that has shown moderate correlations with the Block 1998 Food Frequency Questionnaire regarding intake of specific food groups.[54] For the current study, diet quality was calculated as the sum score of each of the 13 scored REAPS items, ranging from 13 (worst) to 39 (best).

Current, typical sleep duration was recorded via reports of typical times going to sleep and waking up on weekdays and weeknights, separately. The sleep duration variable was calculated as the average number of hours slept per night weighted by weekday and weekend nights. Sleep duration was then categorised into three groups based on the WHO's Study on Global Ageing and Adult Health[46 47] : 'Low' (<6 hours per night), 'Intermediate' (6–9 hours per night), 'High' (>9 hours per night). Low and High sleep groups were dummy-coded so that each group was compared with the Intermediate group as the referent category (eg,<6 hours per night vs 6–9 hours per night).

Covariates

Covariates were selected a priori based on previous literature.[8 15 31 55–60] Demographic covariates included: age; body mass index; race/ethnicity (dichotomous; white/non-Hispanic vs

Br J Sports Med: first published as 10.1136/bjsports-2020-103400 on 4 January 2021. Downloaded from http://bjsm.bmj.com/ on January 5, 2021 at Nationwide Children's Hospital. Protected by copyright.

Table 2 PROMIS and Neuro-QoL outcome measures.

PROMIS Cognitive (COG) Function In the past 7 days…	Never	Rarely (once)	Sometimes (two or three times)	Often (about once a day)	Very often (several times a day	Composite T-score Mean (SD)	Composite T-score Median (IQR)
My thinking has been slow.	173 (9.7)	208 (11.7)	470 (26.3)	509 (28.5)	409 (22.9)	45.3 (10.2)	45.5 (39.2–51.6)
It has seemed like my brain was not working as well as usual.	152 (8.5)	211 (11.8)	437 (24.5)	487 (27.3)	484 (27.1)		
I have had to work harder than usual to keep track of what I was doing.	197 (11.0)	255 (14.3)	373 (20.9)	431 (24.2)	512 (28.7)		
I have had trouble shifting back and forth between different activities that require thinking.	195 (10.9)	212 (11.9)	388 (21.7)	430 (24.1)	543 (30.4)		
PROMIS Depressive (DEP) Symptoms In the past 7 days…	Never	Rarely	Sometimes	Often	Always	Composite T-score Mean (SD)	Composite T-score Median (IQR)
I felt worthless.	1057 (59.2)	310 (17.4)	259 (14.5)	103 (5.8)	32 (1.8)	50.8 (9.9)	49.0 (41.0–57.3)
I felt helpless.	1023 (57.3)	330 (18.5)	257 (14.4)	116 (6.5)	32 (1.8)		
I felt depressed.	811 (45.5)	348 (19.5)	380 (21.3)	166 (9.3)	58 (3.3)		
I felt hopeless.	1058 (59.3)	317 (17.8)	232 (13.0)	109 (6.1)	32 (1.8)		
PROMIS Anxiety (ANX) Symptoms In the past 7 days…	Never	Rarely	Sometimes	Often	Always	Composite T-score Mean (SD)	Composite T-score median (IQR)
I felt fearful.	885 (49.6)	380 (21.3)	337 (18.9)	119 (6.7)	35 (2.0)	53.5 (10.5)	53.7 (40.3–61.4)
I found it hard to focus on anything other than my anxiety.	810 (45.4)	403 (22.6)	335 (18.8)	171 (9.6)	49 (2.7)		
My worries overwhelmed me.	755 (42.3)	430 (24.1)	350 (19.6)	170 (9.5)	52 (2.9)		
I felt uneasy.	636 (35.7)	437 (24.5)	440 (24.7)	202 (11.3)	44 (2.5)		
Neuro-QoL Emotional-Behavioral Dyscontrol (EBD) In the past 7 days…	Never	Rarely	Sometimes	Often	Always	Composite T-score Mean (SD)	Composite T-score Median (IQR)
I had trouble controlling my temper.	547 (30.7)	565 (31.7)	461 (25.8)	142 (8.0)	53 (3.0)	51.1 (10.8)	52.0 (43.7–58.3)
It was hard to control my behavior.	765 (42.9)	582 (32.6)	299 (16.8)	93 (5.2)	28 (1.6)		
I said or did things without thinking.	573 (32.1)	574 (32.2)	431 (24.2)	150 (8.4)	36 (2.0)		
I got impatient with other people.	288 (16.1)	456 (25.6)	599 (33.6)	333 (18.7)	90 (5.0)		
I was irritable around other people.	386 (21.6)	577 (32.3)	491 (27.5)	245 (13.7)	69 (3.9)		
I was bothered by little things.	374 (21.0)	548 (30.7)	528 (29.6)	231 (12.9)	84 (4.7)		
I became easily upset.	533 (29.9)	588 (33.0)	394 (22.1)	180 (10.1)	67 (3.8)		
I was in conflict with others.	581 (32.6)	727 (40.8)	319 (17.9)	97 (5.4)	34 (1.9)		

This table presents proportions of each response as n (per cent) as well as composite T-score distributions of each outcome measure for all respondents who completed each individual measure (Patient-Reported Outcomes Measurement Information System (PROMIS) Cognitive Function (COG) n=1749; PROMIS Depressive Symptoms (DEP) n=1737; PROMIS Anxiety Symptoms (ANX) n=1728; and Neuro-QoL Emotional-Behavioral Dyscontrol (EBD) n=1770). Participant responses to individual outcome measure items were not imputed in the current study. Outcome measure T-scores were imputed for those missing data prior to regression analyses as described elsewhere in this paper and its supplement. IQR range, expressed as 25th percentile–75th percentile.

person of colour); current marital status (dichotomous; married/cohabitating vs not cohabitating); highest education attained (ordinal; less than a bachelor's degree, bachelor's degree, post-graduate degree); and total years of football played (inclusive of prehigh school, high school, college and professional settings). Lifetime physician diagnosis of osteoarthritis or degenerative arthritis (dichotomous; yes/no) was also included as it has been previously shown to be associated with lower quality of life and mood-related symptoms in former elite athletes.[55] Behavioural covariates included: current frequency of alcohol intake (ordinal; 'never', 'Monthly or less', '2–4 times a month', '2–3 times a week' or '4 or more times a week') and illicit drug usage over the prior 3 months (dichotomous; yes/no). Finally, typical experience of pain over the past 7 days was reported on an 11-point Likert Scale (0=no pain to 10=worst pain) as part of the PROMIS-29 Profile V.2.0.[49]

Data processing and statistical analyses

All analyses were performed with SPSS V.25.0 (Armonk, New York, USA). Separate multivariable linear regression models were fit for COG, DEP, ANX and EBD. Missingness observed within the analysis data set was determined not to be missing completely at random, and consequently, multiple imputation was used while fitting these models. Details on this process are included in the supplement. To fit the regression models using pooled data, standardised values for each predictor and outcome (Z-scores with a mean of 0 and SD of 1) were employed. Normality and heteroscedasticity (homogeneity of variance) of residuals were

Br J Sports Med: first published as 10.1136/bjsports-2020-103400 on 4 January 2021. Downloaded from http://bjsm.bmj.com/ on January 5, 2021 at Nationwide Children's Hospital. Protected by copyright.

visually inspected with P–P and spread-level plots, respectively, for each model and variance inflation factors were examined to probe multicollinearity in order to ensure that none of the variables yielded values above the recommended cut-off point of 10.[61] Adjusted R^2 and standardised beta coefficients (β) for each predictor were calculated for each model. Standardised β values were considered statistically significant if their 95% CIs did not include zero. Self-reported SRC history and health-promoting behaviours (exercise frequency, diet quality and sleep duration) were included for each outcome alongside all the selected covariates described above. Analyses were additionally performed both with and without participants who reported dementia-related diagnoses (eg, Alzheimer's disease); however, the results were nearly identical, and these models were therefore not considered further in the present study (online supplemental table 1). Additionally, the summed relative statistical effects (|β|) of exercise frequency, diet quality and Low Sleep were aggregated into a single 'Health-Promoting Habits' effect for comparison with the statistical effects of self-reported SRC on each outcome.

RESULTS

A total of 1784 participants responded to our survey and were subsequently included in this study (figure 1). Most respondents reported that they sustained more than three lifetime SRCs with one-quarter of the sample reporting 10 or more. Approximately one-half of former players indicated that difficulties with cognition occurred 'often' or 'very often.' Overall, self-reported COG, DEP, ANX and EBD approximated population norms (table 2).[48 49 62]

Multivariable linear regressions

Overall, the multivariable regression models explained approximately 33%–38% of the variance (ie, adjusted R^2) associated with each outcome (table 3). Following covariate adjustment, a greater history of self-reported SRC was significantly associated with worse self-reported COG, DEP, ANX and EBD (table 3 and figure 2). More frequent exercise and better diet quality were significantly associated with better outcomes in each domain and sleeping less than 6 hours per night (compared with 6–9 hours) was associated with worse DEP, ANX and EBD, but not COG. Sleeping more than 9 hours per night was not significantly associated with any outcome when compared with Intermediate sleep (6–9 hours). Additionally, the statistical effect size (β) of 'Health-Promoting Habits' was slightly smaller than that of self-reported SRC on COG, and was relatively larger than self-reported SRC on the DEP, ANX and EBD outcomes (figure 2). Corresponding β values for covariates in these regression models are presented in online supplemental table 2.

DISCUSSION

This study, with a large sample of former NFL players, represents an important step toward identifying modifiable, health-promoting factors associated with daily functioning in former athletes. Explanatory variables individually had small to moderate statistical effects on self-reported functioning, and the combination of these variables, alongside important covariates, explained a substantial proportion of variance (ie, adjusted R^2) for each outcome. Reporting a higher number of lifetime SRCs was significantly associated with worse cognitive function, depressive and anxiety symptoms, and emotional–behavioural dyscontrol. Conversely, significant relationships between health-promoting behaviours and each of these outcomes were identified such that more frequent exercise, better diet quality and sleeping six or

Table 3 Regression model predictors of interest

		COG			DEP			ANX			EBD		
Predictor	Value	β	95% CI	P value	β	95% CI	P value	β	95% CI	P value	β	95% CI	P value
SRC history	1-unit increase in category	−0.217*	−0.257 to −0.177	<0.001	0.144*	0.103 to 0.184	<0.001	0.156*	0.117 to 0.195	<0.001	0.148*	0.108 to 0.189	<0.001
Exercise frequency	Increase of 1 session/week	.075*	.031 to 0.120	0.001	−0.082*	−0.127 to −0.036	0.001	−0.088*	−0.132 to −0.043	<0.001	−0.064*	−0.106 to −0.022	0.003
Diet quality	1-point increase in REAPS sum score	.062*	.015 to 0.109	0.011	−0.075*	−0.121 to −0.029	0.002	−0.057*	−0.101 to −0.014	0.010	−0.086*	−0.129 to −0.043	<0.001
Low sleep	Sleeping less than 6 hours versus 6–9 hours	−0.036	−0.077 to 0.004	0.080	0.084*	.040 to 0.127	<0.001	.061*	.019 to 0.103	0.004	0.093*	0.051 to 0.135	<0.001
High sleep	Sleeping more than 9 hours versus 6–9 hours	0.002	−0.037 to 0.042	0.909	0.030	−0.012 to 0.073	0.161	−0.005	−0.046 to 0.036	0.813	−0.004	−0.044 to 0.035	0.834

Standardised beta-weights (β) and 95% CIs are presented for each predictor of interest as calculated from the standardised values (Z-scores) from pooled data following the five iterations of multiple imputation (see supplement for a description of the imputation methods employed). Adjusted R^2 values are presented as the range of values estimated for each model that was fit from the five multiple imputation iterations: PROMIS Cognitive Function (COG), 0.354 to 0.362; PROMIS-29 Depressive Symptoms (DEP), 0.344 to 0.348; PROMIS-29 Anxiety (ANX), 0.372 to 0.382; and Neuro QoL Emotional–Behavioral Dyscontrol (EBD), 0.333 to 0.344. Self-reported sport-related concussion history (SRC History) was included as an ordinal variable according to groups (0 concussions, 1–2 concussions, 3–5 concussions, 6–9 concussions and 10 or more concussions). Current exercise frequency was the sum of the typical number of moderate-to-vigorous cardiovascular exercise and resistance training sessions completed per week. Current diet quality was the REAPS sum score (higher score indicated better diet quality). Current sleep: low sleep β represents a comparison of sleeping less than 6 hours per night to sleeping 6–9 hours per night. High sleep β represents a comparison of sleeping longer than 9 hours per night to sleeping 6–9 hours per night. Covariates in each model included: age, body mass index, race/ethnicity, relationship status, highest education attained, total years playing football (youth through professional), pain (Likert scaled from 0 'no pain' to 10 'worst pain'), osteoarthritis/degenerative arthritis diagnosis (dichotomous; yes vs no), frequency of alcohol consumption (ordinal) and illicit drug use within the past 3 months (dichotomous; yes vs no).

*Statistically significant association between covariate and outcome.

ANX, PROMIS-29 Anxiety; COG, PROMIS Cognitive Function; DEP, PROMIS-29 Depressive Symptoms; EBD, Neuro QoL Emotional–Behavioral Dyscontrol; SRC, sport-related concussion.

Br J Sports Med: first published as 10.1136/bjsports-2020-103400 on 4 January 2021. Downloaded from http://bjsm.bmj.com/ on January 5, 2021 at Nationwide Children's Hospital. Protected by copyright.

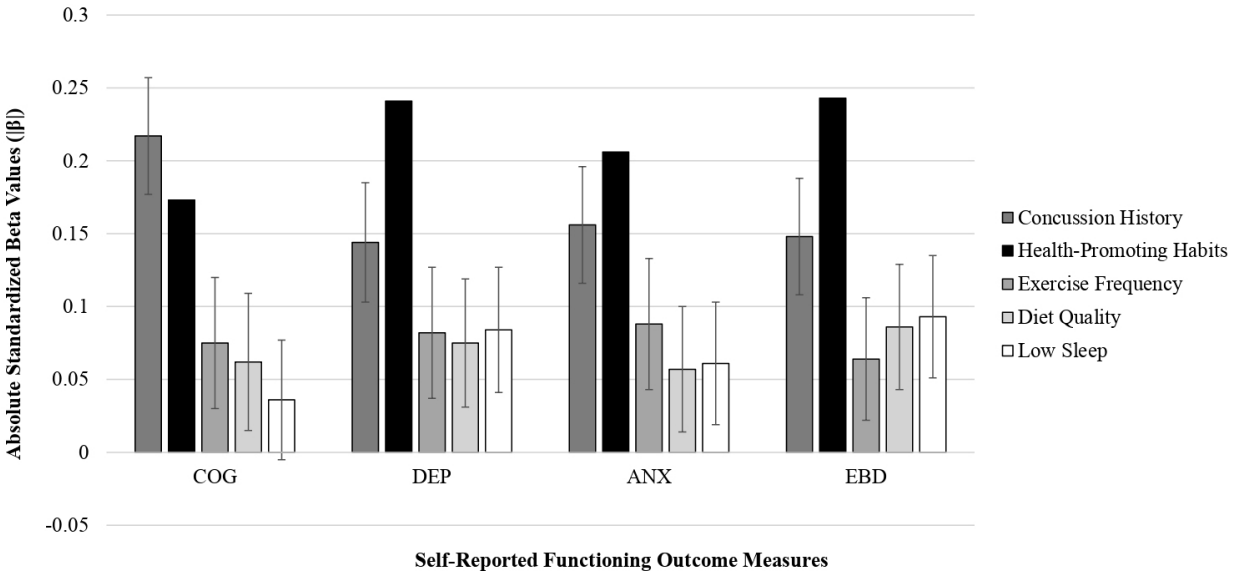

Figure 2 Absolute standardised beta values (|β|) for predictors of Cognitive Function, Depression Symptoms, Anxiety Symptoms and Emotional–Behavioral Dyscontrol from multivariable linear regression models. Four separate multivariable linear regression models were created, one for each outcome (Cognitive Function, Depression Symptoms, Anxiety Symptoms and Emotional–Behavioral Dyscontrol). The following were included as predictors of interest: self-reported sport-related concussion history (Concussion History) was included as an ordinal variable according to groups (0 concussions, 1–2 concussions, 3–5 concussions, 6–9 concussions and 10 or more concussions). Health-Promoting Habits were the sum of the absolute β values for Exercise Frequency, Diet Quality and Low Sleep. Exercise frequency was the sum of the typical number of moderate-to-vigorous cardiovascular exercise and resistance training sessions completed per week. Diet quality was the Rapid Eating Assessment for Participants—Shortened Version sum score (higher score indicated better diet quality). Low Sleep β represents a comparison of sleeping less than 6 hours per night to sleeping 6–9 hours per night. Covariates in each model included: age, body mass index, race/ethnicity, relationship status, highest education attained, total years playing football (youth through professional), pain (Likert scaled from 0 'no pain' to 10 'worst pain'), osteoarthritis/degenerative arthritis diagnosis (dichotomous; yes vs no), frequency of alcohol use (ordinal) and illicit drug use within the past 3 months (dichotomous; yes vs no). Frequency of alcohol consumption was not included as a covariate as it had no bivariate association with any of the outcomes. Vertical bars represent 95% CIs for the absolute β values; no CI was estimated for Health-Promoting Habits as it was aggregated from three separate predictors. ANX=PROMIS-29 Anxiety; COG=PROMIS Cognitive Function; DEP=PROMIS-29 Depressive Symptoms; EBD=Neuro QoL Emotional–Behavioral Dyscontrol.

more hours per night were each related to better self-reported functioning. The combined magnitude of beneficial effects (β values) observed for these behaviours was relatively similar to or greater than the deleterious effects of increasing SRC history group (figure 2). These findings suggest that even minor changes to health-promoting behaviours may improve cognitive and mental health. As such, these modifiable behaviours are potential targets for therapeutic interventions designed to improve the health and quality of life of former players.

Self-reported concussion history

The observation that self-reported SRC history was related to each outcome is similar to previous work with self-reported cognitive function and mood-related symptoms.[8 10 11 14 15] For instance, Guskiewicz *et al*[8] reported worse SF-36 Mental Component Scale scores in former NFL players with three or more prior concussions when compared with those with fewer. Further, lifetime prevalence of depression diagnosis along with odds of self-reporting moderate or severe symptoms of depression appears significantly higher in former football players reporting three or more prior concussions when compared with those reporting fewer.[10 15] Unlike these studies, SRC history was grouped into five categories in the current study—three of these categories included participants that would have been sorted into the '3 or more' category in prior studies,[8 10 15] allowing for more granularity in interpreting the possible effects of multiple prior SRCs. Still, the current findings that self-reporting more SRCs was significantly associated with worse self-reported cognitive and mood-related functioning is consistent with this prior work (table 3, online supplemental table 2).

Health-promoting behaviours

A unique component of this study with former NFL players was the investigation of current exercise, diet and sleep habits as possible promoters of health-related functions. The extant literature regarding the benefits of regular exercise on cognitive function, mood-related symptoms and brain health is robust—but less available in former American football players.[26 28 33 36 42 45 63] Previous survey studies with former NFL players have shown that self-reported cognition-related problems and moderate to severe depressive symptoms were associated with low levels of physical activity and loss of fitness.[28 33] It is unclear, however, whether the depressive symptoms endorsed by players in those studies (as well as the current study) were due to the psychosocial experience during and prior to transition from a sport they had played for years, decreased physical activity, changes in self-esteem or identity, an underlying neurobiological process related to depression symptoms, or some complex mixture thereof. In this study, participation in moderate-to-vigorous aerobic exercise and/or resistance training was significantly associated with better scores on measures of cognition, depressive symptoms, anxiety and behavioural dyscontrol. Compared with the observed deleterious statistical effect (β value) of increasing one concussion history category (eg, 1–2 lifetime SRCs to 3–5 SRCs) for each

Br J Sports Med: first published as 10.1136/bjsports-2020-103400 on 4 January 2021. Downloaded from http://bjsm.bmj.com/ on January 5, 2021 at Nationwide Children's Hospital. Protected by copyright.

outcome, the present findings suggest that two or three additional exercise sessions per week may have a favourable effect with comparable magnitude.

Better diet quality (higher REAPS score) was significantly associated with better COG, DEP, ANX and EBD outcomes. Statistical interpretation of this measure in the current study infers that higher quality food choices may result in measurable improvements in daily function. For example, to improve REAPS score by one point, the unit of measure in this study, one could increase the frequency that they consume more than two servings of vegetables in a day, or decrease their frequency of eating fried foods.[54] In a related study with former football players, higher fat and cholesterol intake were significantly correlated with greater self-reported cognitive difficulties, suggesting that diet choices are associated with daily functioning.[26] Like exercise, dietary changes may have a dose–response relationship with self-reported functioning, for better or worse, warranting further longitudinal investigation with more sophisticated assessments of dietary intake behaviours. Furthermore, dietary recommendations are tangible interventions with the potential to improve daily functioning in this population.

The relationships between typical nightly sleep duration and the outcomes of interest were consistent with previous work that identified durations of sleep of greater than 6 hours per night as being preferable to fewer hours.[46 47] Sleeping less than 6 hours per night has been related to worse cognitive functioning and both structural and functional brain health (eg, increased atrophy and functional connectivity changes over time, respectively) across the lifespan.[37 38 46 47 64] In former American football players, those reporting 'trouble sleeping' have exhibited higher odds of reporting moderate to severe depression symptoms.[33] In the present study, being in the 'low sleep' group was significantly associated with worse DEP, ANX and EBD, but not COG (the upper limit of the 95% CI for β barely crossed zero). It's important to consider that sleep duration may not equate to quality of sleep, and a more comprehensive assessment of sleep is needed to fully understand the relationships among sleep and self-reported function, as well as to identify targets for specific sleep-related interventions. Although sleep quality and napping habits were not included in the present study, our results suggest that sleeping at least 6 hours per night is preferable.

Considering the beneficial effects suggested by the data in this study, former players and healthcare providers should discuss the potential for incorporating exercise-related, diet-related and sleep-related behaviours as preventative and therapeutic agents for health-related function. Future research should examine how these behaviours relate to self-reported and objectively measured markers of health through longitudinal investigations with former athletes.

Limitations

First, it should be acknowledged that the response rate for this survey was relatively low. However, an exact number could not be calculated as the authors did not have access to all the necessary data, such as invalid email addresses. Relatedly, we cannot rule out the possibility that the sample of survey respondents was biased in some way (eg, representing a cohort of individuals that are more eager to participate in research studies because of current health-related issues) and that participants who chose not to respond to this survey may not be adequately represented by those who did participate in this study. Second, some of the self-reported predictors and dependent outcomes are limited as they were not corroborated with objective assessments. For example, cognitive functioning was self-reported and was not assessed via neuropsychological testing, and medical chart reviews were not performed to corroborate diagnoses (eg, SRC). As such, we cannot objectively confirm what respondents reported. In addition, respondents were asked to report concussion that occurred years, even decades, earlier, while aware of any potential difficulties they now have, making these responses subject to recall bias. It has been speculated that recalling SRC history from years or decades past may change over time and in relation to changes in health status or understanding of concussion.[65] Additionally, it is impossible to discern the directionality or causal nature of relationships observed in a cross-sectional study such as this one. Longitudinal investigations are imperative to determine the natural history and modifiers of health and function over time in former athletes. Finally, predictor and covariate selection for the current study was based on *a-priori* prioritisation of factors that were suspected to have a meaningful influence on COG, DEP, ANX and EBD, but these variables do not represent an exhaustive list and are consequently vulnerable to residual confounding.

What are the new findings?

- Sport-related concussion history, exercise frequency, diet quality and sleep duration appear to have measurable individual and aggregate associations with self-reported measures of cognitive, mood-related and behavioural functioning.

How might it impact on clinical practice in the near future?

- Although concussion history is unchangeable in former athletes, exercise, diet and sleep are modifiable health-promoting behaviours that warrant future investigation as therapeutic and preventative clinical interventions to improve health and quality of life in former American football players.

CONCLUSION

Several variables appear to be associated with the health status of former NFL players. Although SRC history is non-modifiable in former NFL players, the aggregate statistical effects of increasing exercise frequency, making dietary improvements and obtaining adequate sleep duration could be sufficient to impart benefits to self-reported functioning and should be explored as therapeutic targets for health-related outcomes in former players.

Twitter Samuel R Walton @SammoWalton and Benjamin L Brett @BenjaminBrett1

Acknowledgements The authors are grateful to Mrs Candice Goerger, Ms Hope Campbell, and Ms. Caprice Hunt for their invaluable efforts in the administration, coordination, and execution of the NFL-LONG study. Special thanks are in order for members of the NFL Alumni Association and Legends community for their assistance in recruitment of study participants. Further, we would like to thank each of the clinicians, researchers, educators, and former players who helped us adapt and develop the current general health survey.

Contributors All authors agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. Specific author contributions are as follows: SRW, ZYK, BLB and AC: creation and distribution of the study survey, management of study database, established research hypotheses, planned analyses of the data, interpretation of study findings, manuscript preparation and revision, final approval of the submitted manuscript. JDDF, AES-R, LS, RJE, MM, WPM III and KMG: creation and distribution of the study survey, interpretation of study findings,

Br J Sports Med: first published as 10.1136/bjsports-2020-103400 on 4 January 2021. Downloaded from http://bjsm.bmj.com/ on January 5, 2021 at Nationwide Children's Hospital. Protected by copyright.

manuscript preparation and revision, final approval of the submitted manuscript. Other contributors to the project are as follows (included in acknowledgements): Mrs CG, Project Administrator: Assisted in the creation and distribution of the survey instrument as well as collection and management of the survey responses. Additionally, she was responsible for overseeing the project administration and execution. Ms HC, Project Coordinator (Medical College of Wisconsin): Assisted in the creation and distribution of the survey instrument as well as collection and management of the survey responses. Additionally, she assisted with oversight of the project administration and execution. Ms. Caprice Hunt, Project Coordinator (University of North Carolina at Chapel Hill): Assisted in the creation and distribution of the survey, oversight of the project administration, and execution of the study.

Funding Funding for this research project comes from the National Football League and Boston Children's Hospital.

Competing interests BLB acknowledges support from the National Institute of Neurological Disorders and Stroke under the National Institutes of Health under the award NO L301L30NS113158-01. RJE is a paid consultant for the NHL and co-chair of the NHL/NHLPA Concussion Subcommittee. He is also a paid consultant for Major League Soccer and Princeton University Athletic Medicine and occasionally provides expert testimony in matters related to MTBI and sports concussion. WPM receives royalties from 1) ABC-Clio publishing for the sale of his books, *Kids, Sports, and Concussion: A guide for coaches and parents,* and *Concussions;* 2) Springer International for the book *Head and Neck Injuries in Young Athlete* and 3) Wolters Kluwer for working as an author for *UpToDate*. His research is funded, in part, by philanthropic support from the National Hockey League Alumni Association through the Corey C. Griffin Pro-Am Tournament and a grant from the National Football League.

Patient consent for publication Not required.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement The data from this survey study are available upon reasonable request. Inquiries may be sent to Dr ZYK.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

ORCID iDs
Samuel R Walton http://orcid.org/0000-0003-2647-5701
Zachary Y Kerr http://orcid.org/0000-0003-1423-1259
Benjamin L Brett http://orcid.org/0000-0003-2849-4658
Ruben J Echemendia http://orcid.org/0000-0001-6116-8462
Michael McCrea http://orcid.org/0000-0001-9791-9475

REFERENCES

1 Kerr ZY, Chandran A, Nedimyer AK, *et al*. Concussion incidence and trends in 20 high school sports. *Pediatrics* 2019;144:e20192180.
2 Zuckerman SL, Kerr ZY, Yengo-Kahn A, *et al*. Epidemiology of sports-related concussion in NCAA athletes from 2009-2010 to 2013-2014: incidence, recurrence, and mechanisms. *Am J Sports Med* 2015;43:2654–62.
3 Manley G, Gardner AJ, Schneider KJ, *et al*. A systematic review of potential long-term effects of sport-related concussion. *Br J Sports Med* 2017;51:969–77.
4 Montenigro PH, Alosco ML, Martin BM, *et al*. Cumulative head impact exposure predicts later-life depression, apathy, executive dysfunction, and cognitive impairment in former high school and College football players. *J Neurotrauma* 2017;34:328–40.
5 Fields L, Didehbani N, Hart J, *et al*. No linear association between number of concussions or years played and cognitive outcomes in retired NFL players. *Arch Clin Neuropsychol* 2020;35:233–9.
6 Ruiter KI, Boshra R, Doughty M, *et al*. Disruption of function: neurophysiological markers of cognitive deficits in retired football players. *Clin Neurophysiol* 2019;130:111–21.
7 Clark MD, Varangis EML, Champagne AA, *et al*. Effects of career duration, concussion history, and playing position on white matter microstructure and functional neural recruitment in former college and professional football athletes. *Radiology* 2018;286:967–77.
8 Guskiewicz KM, Marshall SW, Bailes J, *et al*. Association between recurrent concussion and late-life cognitive impairment in retired professional football players. *Neurosurgery* 2005;57:719–26.
9 Ford JH, Giovanello KS, Guskiewicz KM. Episodic memory in former professional football players with a history of concussion: an event-related functional neuroimaging study. *J Neurotrauma* 2013;30:1683–701.
10 Kerr ZY, Thomas LC, Simon JE, *et al*. Association between history of multiple concussions and health outcomes among former college football players: 15-year follow-up from the NCAA concussion study (1999-2001). *Am J Sports Med* 2018;46:1733–41.
11 Tremblay S, Henry LC, Bedetti C, *et al*. Diffuse white matter tract abnormalities in clinically normal ageing retired athletes with a history of sports-related concussions. *Brain* 2014;137:2997–3011.
12 Strain JF, Womack KB, Didehbani N, *et al*. Imaging correlates of memory and concussion history in retired National football League athletes. *JAMA Neurol* 2015;72:773.
13 Roberts AL, Pascual-Leone A, Speizer FE, *et al*. Exposure to American football and neuropsychiatric health in former national football League players: findings from the football players health study. *Am J Sports Med* 2019;47:2871–80.
14 Kerr ZY, Evenson KR, Rosamond WD, *et al*. Association between concussion and mental health in former collegiate athletes. *Inj Epidemiol* 2014;1:28.
15 Guskiewicz KM, Marshall SW, Bailes J, *et al*. Recurrent concussion and risk of depression in retired professional football players. *Med Sci Sports Exerc* 2007;39:903–9.
16 Bieniek KF, Blessing MM, Heckman MG, *et al*. Association between contact sports participation and chronic traumatic encephalopathy: a retrospective cohort study. *Brain Pathol* 2020;30:bpa.12757:63–74.
17 Kerr ZY, Marshall SW, Harding HP, *et al*. Nine-Year risk of depression diagnosis increases with increasing self-reported concussions in retired professional football players. *Am J Sports Med* 2012;40:2206–12.
18 Iverson GL, Terry DP, Luz M, *et al*. Anger and depression in middle-aged men: implications for a clinical diagnosis of chronic traumatic encephalopathy. *J Neuropsychiatry Clin Neurosci* 2019;31:328–36.
19 Strain J, Didehbani N, Cullum CM, *et al*. Depressive symptoms and white matter dysfunction in retired NFL players with concussion history. *Neurology* 2013;81:25–32.
20 Willer BS, Tiso MR, Haider MN, *et al*. Evaluation of executive function and mental health in retired contact sport athletes. *J Head Trauma Rehabil* 2018;33:E9–15.
21 Misquitta K, Dadar M, Tarazi A, *et al*. The relationship between brain atrophy and cognitive-behavioural symptoms in retired Canadian football players with multiple concussions. *Neuroimage Clin* 2018;19:551–8.
22 Alosco ML, Koerte IK, Tripodis Y, *et al*. White matter signal abnormalities in former national football League players. *Alzheimers Dement* 2018;10:56–65.
23 Lehman EJ, Hein MJ, Gersic CM. Suicide mortality among retired National football League players who played 5 or more seasons. *Am J Sports Med* 2016;44:2486–91.
24 Webner D, Iverson GL. Suicide in professional American football players in the past 95 years. *Brain Inj* 2016;30:1718–21.
25 Seichepine DR, Stamm JM, Daneshvar DH, *et al*. Profile of self-reported problems with executive functioning in college and professional football players. *J Neurotrauma* 2013;30:1299–304.
26 Hinton PS, Johnstone B, Blaine E, *et al*. Effects of current exercise and diet on late-life cognitive health of former college football players. *Phys Sportsmed* 2011;39:11–22.
27 Roberts AL, Zafonte RD, Speizer FE, *et al*. Modifiable risk factors for poor cognitive function in former American-style football players: findings from the Harvard football players health study. *J Neurotrauma* 2020:neu.2020.7070.
28 Plessow F, Pascual-Leone A, McCracken CM, *et al*. Self-Reported cognitive function and mental health diagnoses among former professional American-style football players. *J Neurotrauma* 2020;37:neu.2019.6661
29 Baker JG, Leddy JJ, Hinds AL, *et al*. An exploratory study of mild cognitive impairment of retired professional contact sport athletes. *J Head Trauma Rehabil* 2018;33:E16–23.
30 Didehbani N, Munro Cullum C, Mansinghani S, *et al*. Depressive symptoms and concussions in aging retired NFL players. *Arch Clin Neuropsychol* 2013;28:418–24.
31 Alosco ML, Mez J, Tripodis Y. Age of first exposure to tackle football and chronic traumatic encephalopathy: age of first exposure to football and CTE. *Ann Neurol* 2018;83:886–901.
32 Brett BL, Mummareddy N, Kuhn AW, *et al*. The relationship between prior concussions and depression is modified by somatic symptomatology in retired NFL athletes. *J Neuropsychiatry Clin Neurosci* 2019;31:17–24.
33 Schwenk TL, Gorenflo DW, Dopp RR, *et al*. Depression and pain in retired professional football players. *Med Sci Sports Exerc* 2007;39:599–605.
34 Willeumier K, Taylor DV, Amen DG. Elevated body mass in national football League players linked to cognitive impairment and decreased prefrontal cortex and temporal pole activity. *Transl Psychiatry* 2012;2:e68.
35 Kelly DF, Chaloner C, Evans D, *et al*. Prevalence of pituitary hormone dysfunction, metabolic syndrome, and impaired quality of life in retired professional football players: a prospective study. *J Neurotrauma* 2014;31:1161–71.
36 Erickson KI, Voss MW, Prakash RS, *et al*. Exercise training increases size of hippocampus and improves memory. *Proc Natl Acad Sci U S A* 2011;108:3017–22.
37 Richards A, Inslicht SS, Metzler TJ, *et al*. Sleep and cognitive performance from teens to old age: more is not better. *Sleep* 2017;40.
38 JC L, Loh KK, Zheng H. Sleep duration and age-related changes in brain structure and cognitive performance. *Sleep* 2014;37:821.

Br J Sports Med: first published as 10.1136/bjsports-2020-103400 on 4 January 2021. Downloaded from http://bjsm.bmj.com/ on January 5, 2021 at Nationwide Children's Hospital. Protected by copyright.

39 Allès B, Samieri C, Jutand M-A, *et al*. Nutrient patterns, cognitive function, and decline in older persons: results from the Three-City and nuage studies. *Nutrients* 2019;11:nu11081808. doi:10.3390/nu11081808
40 Brinkworth GD. Long-Term effects of a very low-carbohydrate diet and a low-fat diet on mood and cognitive function. *Arch Intern Med* 1873;2009:169.
41 Lourida I, Soni M, Thompson-Coon J, *et al*. Mediterranean diet, cognitive function, and dementia: a systematic review. *Epidemiology* 2013;24:479–89.
42 Adamson BC, Ensari I, Motl RW. Effect of exercise on depressive symptoms in adults with neurologic disorders: a systematic review and meta-analysis. *Arch Phys Med Rehabil* 2015;96:1329–38.
43 Barbosa Filho VC, Oppa DF, Mota J, *et al*. Predictors of health-related quality of life among Brazilian former athletes. *Rev Andal Med Deport* 2018;11:23–9.
44 Batista C, Soares JM. Are former elite athletes more protected against metabolic syndrome? *J Cardiol* 2013;61:440–5.
45 Hamer M, Chida Y. Physical activity and risk of neurodegenerative disease: a systematic review of prospective evidence. *Psychol Med* 2009;39:3–11.
46 Gildner TE, Salinas-Rodríguez A, Manrique-Espinoza B, *et al*. Does poor sleep impair cognition during aging? longitudinal associations between changes in sleep duration and cognitive performance among older Mexican adults. *Arch Gerontol Geriatr* 2019;83:161–8.
47 Gildner TE, Liebert MA, Kowal P, *et al*. Associations between sleep duration, sleep quality, and cognitive test performance among older adults from six middle income countries: results from the study on global ageing and adult health (SAGE). *J Clin Sleep Med* 2014;10:613–21.
48 Lai J-S, Wagner LI, Jacobsen PB. Self-Reported cognitive concerns and abilities: two sides of one coin?: cognitive concerns versus cognitive abilities. *Psychooncology* 2014;23:1133–41.
49 Hays RD, Spritzer KL, Schalet BD, *et al*. PROMIS®-29 v2.0 profile physical and mental health summary scores. *Qual Life Res* 2018;27:1885–91.
50 Gershon RC, Lai JS, Bode R, *et al*. Neuro-QOL: quality of life item banks for adults with neurological disorders: item development and calibrations based upon clinical and general population testing. *Qual Life Res* 2012;21:475–86.
51 Reeve BB, Hays RD, Bjorner JB, *et al*. Psychometric evaluation and calibration of health-related quality of life item banks: plans for the patient-reported outcomes measurement information system (PROMIS). *Med Care* 2007;45:S22–31.
52 McCrea M, Hammeke T, Olsen G, *et al*. Unreported concussion in high school football players: implications for prevention. *Clin J Sport Med* 2004;14:13–17.
53 McCrory P, Meeuwisse W, Dvořák J, *et al*. Consensus statement on concussion in sport-the 5th international conference on concussion in sport held in Berlin, October 2016. *Br J Sports Med* 2017;51:838–47.
54 Segal-Isaacson CJ, Wylie-Rosett J, Gans KM. Validation of a short dietary assessment questionnaire: the rapid eating and activity assessment for participants short version (REAP-S). *Diabetes Educ* 2004;30:774–81.
55 Turner AP, Barlow JH, Heathcote-Elliott C. Long term health impact of playing professional football in the United Kingdom. *Br J Sports Med* 2000;34:332–6.
56 Bellou V, Belbasis L, Tzoulaki I, *et al*. Systematic evaluation of the associations between environmental risk factors and dementia: an umbrella review of systematic reviews and meta-analyses. *Alzheimers Dement* 2017;13:406–18.
57 Matthews KA, Xu W, Gaglioti AH, *et al*. Racial and ethnic estimates of Alzheimer's disease and related dementias in the United States (2015-2060) in adults aged ≥65 years. *Alzheimers Dement* 2019;15:17–24.
58 Norton S, Matthews FE, Barnes DE, *et al*. Potential for primary prevention of Alzheimer's disease: an analysis of population-based data. *Lancet Neurol* 2014;13:788–94.
59 Whitlock EL, Diaz-Ramirez LG, Glymour MM, *et al*. Association between persistent pain and memory decline and dementia in a longitudinal cohort of elders. *JAMA Intern Med* 2017;177:1146.
60 Weber A, Mak SH, Berenbaum F, *et al*. Association between osteoarthritis and increased risk of dementia: a systemic review and meta-analysis. *Medicine* 2019;98:e14355.
61 Hair J, Black W, Babin B. *Multivariate data analysis*. New York, NY: Macmillan, 1995.
62 Cella D, Lai J-S, Nowinski CJ, *et al*. Neuro-QOL: brief measures of health-related quality of life for clinical research in neurology. *Neurology* 2012;78:1860–7.
63 Erickson KI, Gildengers AG, Butters MA. Physical activity and brain plasticity in late adulthood. *Dialogues Clin Neurosci* 2013;15:99–108.
64 Liu Y-R, Fan D-Q, Gui W-J, *et al*. Sleep-Related brain atrophy and disrupted functional connectivity in older adults. *Behav Brain Res* 2018;347:292–9.
65 Kerr ZY, Marshall SW, Guskiewicz KM. Reliability of concussion history in former professional football players. *Med Sci Sports Exerc* 2012;44:377–82.

Br J Sports Med: first published as 10.1136/bjsports-2020-103400 on 4 January 2021. Downloaded from http://bjsm.bmj.com/ on January 5, 2021 at Nationwide Children's Hospital. Protected by copyright.